The Biostat Toolkit

User Guide

A browser-based statistics education platform for clinical and dental researchers

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Contents

1. What Is the Biostat Toolkit?

The Biostat Toolkit is a free, browser-based site built to make statistical concepts and calculations accessible to clinical and dental researchers. It runs entirely in the browser — nothing is installed, no account is required, and no data you enter is transmitted or stored anywhere; every calculation happens locally on your device.

The site is organized into three linked hubs, switchable at any time from the toggle bar at the top of the page:

- Calculators — statistical calculators, organized by category, with formulas, plain-language notation, and (where relevant) charts.
- Learn — reference guides on data types, chart-reading, critical appraisal, and common statistical pitfalls.
- Designs — a quick-reference map from a research question to the study design built for it.

Each hub has its own "finder" wizard for when you're not sure exactly what you're looking for, described in Section 3.

2. Getting Around the Site

2.1 The Sidebar and Search Bar

The left-hand sidebar holds the site logo (click it any time to return to the Calculators home page), a search box, and a mobile navigation toggle on narrow screens. The sidebar's list area is empty until you start typing — as soon as you enter a search term, it fills with matching calculators and guides together, grouped by category and ranked with the best matches first. Clear the box (or click the small × button) to collapse the list back to nothing.

2.2 The Three Hubs

At the top of the Calculators, Learn, and Designs pages sits a shared three-way toggle (Calculators / Learn / Designs) so you can jump between them without going back through the sidebar. Each hub page organizes its contents into collapsible category sections — click a category heading to expand or collapse it.

2.3 Category Sections

On the Calculators home page, every category shows a live count (e.g. "14 available") and expands into a grid of cards, one per calculator, each showing its name, a short formula hint, and a one-line description. The Learn and Designs hubs use the same card layout for guides and designs respectively.

2.4 Breadcrumbs

Every calculator and guide page shows a breadcrumb trail at the top (e.g. Calculators › Epidemiology & Risk ‹calculator name›) so you can jump back to the category or home page in one click.

3. Finding the Right Tool: The Three Finder Wizards

If you know roughly what you need but not the exact name, each hub has its own step-by-step wizard, reachable from the banner near the top of that hub's page:

3.1 Calculator Finder (#wizard)

Answers a short series of questions about your data (analytic goal, variable roles, measurement scale, data structure) and narrows down to the calculator built for that combination. Every step also offers a free-text box — "Or, describe your study in your own words" — where you can type a plain-language description (e.g. "comparing an oral health score between two treatment groups, adjusting for a baseline measurement") and get ranked matches immediately, without answering the questions at all.

3.2 Guide Finder (#learnwizard)

The same step-by-step-or-free-text approach, applied to the Learn Hub's guides instead of calculators — useful when you know the concept you're confused about but not what the relevant guide is called.

3.3 Study Design Finder (#designwizard)

Helps you choose a study design before data collection even begins. Its results mix both a design's critical-appraisal guide and, where one exists, a companion calculator you'll likely need once data exists.

The free-text matching in all three wizards is a keyword match against each item's own description, not true natural-language understanding — it can guess wrong. Read the "why" text under a suggested result before trusting it over the step-by-step path.

4. Using a Calculator Page

Every calculator page follows the same layout:

- Title and description — what the calculator computes.
- Medical Example (where available) — an expandable "Medical Example — when & how to use this" panel with a worked clinical scenario, which updates live as you change the inputs.
- Inputs — the values you enter (a 2×2 table, grouped columns, or a plain grid, depending on the calculator).
- Calculate / Reset buttons — Calculate runs the computation; Reset blanks all fields (it does not restore the original example defaults — reloading the page does that).
- Results — computed values, confidence intervals, and where relevant a chart.
- Formula — the exact formula(s) used, rendered as typeset math.
- Notation — a plain-English legend for every symbol appearing in that calculator's formulas.

Calculators load with example values already filled in and compute automatically on first load, so you always see sample output before entering your own numbers. A handful of calculators — the "Explorer" and simulator tools, such as Power with Graph or the Central Limit Theorem Simulator — use a different, continuously live-updating layout instead of the Calculate-button pattern, since their whole point is watching a chart respond as you drag a slider.

4.1 Exporting a Chart

Any calculator that produces a chart has PNG and JPG download buttons directly beneath it, for pasting into a manuscript, slide, or report.

Full Calculator Index

The Toolkit currently includes 135 calculators across 15 categories. Every calculator listed below is fully available — there are no placeholder or "coming soon" entries. Each opens directly from the Calculators home page, from a search result, or from the Calculator Finder wizard.

Descriptive Statistics (9)

Calculator	What it does
Standard Deviation — Calculated & Visualized	Computes sample and population variance and standard deviation from raw data, then plots the distribution on a normal curve with ±1 SD, ±2 SD, and ±3

Calculator	What it does
	SD bands.
Variance & Standard Deviation	Calculates variance and standard deviation for a sample or population.
Standard Error of the Mean	Computes the standard error of the mean from SD and sample size.
Standard Error of a Proportion	Computes the standard error of a single sample proportion from p and n.
Variance, Standard Deviation & Standard Error of the Mean — Graph	Plots variance, standard deviation, and standard error of the mean together.
RevMan — Finding SD	Derives standard deviations from confidence intervals or standard errors for meta-analysis.
Combining Groups (Mean, SD & N)	Combines 2 to 4 separately reported subgroups (e.g., age bands, sites, or sexes) into one overall mean, SD, and N — as if the raw data itself had been pooled into a single sample.
Interquartile Range (IQR)	Computes Q1, the median, Q3, and the interquartile range from raw data, flags outliers using Tukey's 1.5×IQR fences, and draws a box-and-whisker plot.
Weighted Average / Weighted Score	Combines several scores — exam grades, rubric items, composite quality indices — into one overall score using weights you assign directly, rather than weights derived from sample size.

Epidemiology & Risk (15)

Calculator	What it does
Measures of Association	Computes absolute risk, relative risk, odds ratio, and risk difference — all with 95% confidence intervals — from a 2×2 table of exposure and outcome, plus the number needed to treat for an additional beneficial (NNTB) or harmful (NNTH) outcome, and a two-proportion z-test and chi-square test of significance for the risk difference.
RR/OR & Baseline Risk Explorer	Holds relative risk or odds ratio fixed while you slide the baseline (control-group) risk, showing how the same relative effect translates into a very different absolute risk difference, NNT, and 2×2 table depending on how common the outcome already is.
Nested Case-Control (1:1 Matched)	Computes the matched odds ratio from a 1:1 nested case-control study, where each case is matched to one control sampled from the risk set at that case's event time. Because of that risk-set sampling, the resulting OR directly estimates the incidence rate ratio, not just an approximation of it.
Mantel-Haenszel Stratified 2×2 Analysis	Pools an odds ratio and risk ratio across several 2×2 exposure/outcome tables — one per confounder stratum (site, age band, sex) — using the Mantel-Haenszel method, tests the pooled association, and flags whether the crude (unstratified) and adjusted estimates differ enough to suggest confounding by the stratification variable.
OR to NNTB & NNTH	Converts an odds ratio and baseline control event rate to relative risk, absolute risk difference, and the number needed to treat for an additional beneficial outcome (NNTB, if OR < 1) or an additional harmful outcome (NNTH, if OR > 1). Formerly known as NNT and NNH, respectively.

Calculator	What it does
SE of ln(RR) & ln(OR) — 2×2 Table	Computes the standard error of a difference in proportions, ln(RR), and ln(OR) from a 2×2 table of exposure and outcome counts.
Standard Error of a Rate	Computes the standard error of an incidence rate from the number of events and total person-time.
Standard Error of a Rate Ratio	Computes the standard error of ln(Rate Ratio) and a 95% CI from two groups' event counts and person-time.
Odds Ratio to Risk Ratio	Converts an odds ratio to a risk ratio using the control event rate.
Attributable Fraction (AFe & PAF)	Estimates the fraction of disease attributable to the exposure, both among the exposed (AFe) and across the whole population (Population Attributable Fraction, PAF).
Population Attributable Risk	Quantifies how much of the disease burden would be eliminated by removing the exposure.
Incidence Rate & Rate Ratio	Calculates incidence rates, rate ratio, and their confidence intervals from person-time data.
IPW & ATE	Inverse probability weighting and average treatment effect for observational studies.
Measures of Association — Prediction Intervals	Extends OR and RR estimates with prediction intervals for future studies.
Test for Interaction Between Two Effects	Tests whether two subgroup effect estimates (e.g. RR, OR, HR, or a mean difference) differ significantly from each other, rather than just comparing their individual significance.

T-Tests & Z-Tests (12)

Calculator	What it does
Standard Error of a Mean Difference	Computes the standard error of the difference between two independent sample means, using pooled and Welch (unequal-variance) methods.
Unpaired t-Test (Welch's)	Compares means of two independent groups without assuming equal variances — the recommended default over the classic pooled-variance (Student's) t-test, since it performs about as well when variances are equal and considerably better when they aren't.
Equivalence / Non-Inferiority Test (TOST)	Tests whether the difference between two independent group means is small enough to be considered equivalent (or non-inferior) to a pre-specified margin, using the two one-sided tests (TOST) procedure.
Two-Sample z-Test (Means)	Tests whether two independent population means differ, when both population standard deviations are known.
1-Sample t-Test	Tests whether a sample mean differs from a known population mean, with a configurable significance level and one- or two-tailed testing.
Paired t-Test	Tests the mean difference between paired or matched observations.
1-Sample z-Test	Tests a sample mean against a population mean when σ is known, with a configurable significance level and one- or two-tailed testing.

Calculator	What it does
z-Test Proportions (1-Sample)	Tests whether an observed proportion differs from a hypothesized value.
z-Test Proportions (2-Sample)	Tests whether two independent proportions are equal.
Binomial Hypothesis Test	Exact binomial test for a proportion against a null hypothesis, with graph.
Confidence Interval for a Mean	Computes a t-based confidence interval for a single sample mean, at any confidence level.
Confidence Interval for a Proportion	Computes a z-based (Wald) confidence interval for a single sample proportion, at any confidence level.

Chi-Square & Categorical (7)

Calculator	What it does
Chi-Square 2×2	Chi-square test of independence for a 2×2 contingency table, plus RR, OR, and design-aware interpretation.
Chi-Square Goodness-of-Fit	Tests whether observed frequencies match expected frequencies.
Fisher's Exact Test	Exact test of association for small-sample 2×2 tables.
Fragility Index	Counts how many patients' outcomes would need to change before a statistically significant 2×2 result stops being significant, using Fisher's exact test — along with the Fragility Quotient (FI ÷ N).
McNemar's Test	Tests marginal homogeneity in paired nominal data (before/after or matched pairs).
Cochran's Q Test	Non-parametric test for differences among three or more matched proportions.
Monte Carlo Exact Test (r×c Table)	Estimates an exact p-value for an r×c contingency table by simulating random tables under the same row and column totals — for use when expected cell counts are too low (conventionally <5) to trust the asymptotic chi-square approximation.

Effect Sizes & Agreement (12)

Calculator	What it does
Cohen's d	Quantifies the standardized difference between two group means. Cohen's d is unit-independent, calculated using the pooled standard deviation as the common scale.
Hedges' g	Corrects Cohen's d for small-sample bias and reports a confidence interval — the standardized mean difference typically used as the effect-size input to a meta-analysis.
Cramer's V	Measures the strength of association in contingency tables larger than 2×2.
Phi Coefficient (2×2)	Measures the association between two binary variables in a 2×2 table.
Cohen's Kappa	Measures inter-rater agreement for categorical data, corrected for chance.

Calculator	What it does
Weighted Kappa	Extends Cohen's kappa to ordinal ratings, weighting disagreements by their distance.
Fleiss' Kappa	Chance-corrected agreement among three or more raters on unordered categories.
Intraclass Correlation (ICC)	Assesses reliability and agreement for continuous measurements across raters or occasions — one-way, two-way random, or two-way mixed, single- or average-measures.
Bland-Altman Limits of Agreement	Assesses agreement between two measurement methods from paired data, plotting the mean-difference (Bland-Altman) plot with limits of agreement.
Eta-Squared	Effect size for ANOVA, expressing the proportion of total variance explained by a factor.
Cronbach's Alpha	Measures internal consistency reliability of a multi-item scale or questionnaire.
Is This Difference Important? (MD vs. MID)	Judges whether a mean difference is clinically important by comparing its 95% confidence interval — not just its point estimate — against a minimal important difference (MID), distinguishing a precisely-estimated trivial effect from an interval too wide to be informative.

Correlation & Regression (9)

Calculator	What it does
Standard Error of a Correlation Coefficient	Computes the standard error of a Pearson correlation coefficient from r and sample size, plus a Fisher z-based 95% CI.
Simple Linear Regression	Fits a straight line to data and estimates slope, intercept, R^2, and significance.
Multiple Linear Regression	Fits a linear model with two or more predictors at once, estimating each coefficient, its standard error, R^2, adjusted R^2, and overall significance.
Logistic Regression (2×2)	Estimates the log-odds and OR from a 2×2 table using logistic regression.
Multiple Logistic Regression	Fits a logistic regression model with two or more predictors at once, giving each predictor's adjusted odds ratio (aOR) — holding every other predictor in the model constant — plus its 95% CI, Wald test, and the model's overall likelihood-ratio test.
Multivariable Outcome Predictor	Computes a predicted outcome — or predicted probability — for a new individual from a linear or logistic equation you supply: an intercept plus a coefficient and a value for each predictor (e.g. diet, health behavior, medication use).
Pearson's Correlation	Measures the strength and direction of the linear relationship between two continuous variables.
Spearman's Rank Correlation	Non-parametric measure of monotonic association between two ranked variables.
Kendall's τ	Non-parametric correlation based on the number of concordant vs. discordant pairs.

Power & Sample Size (16)

Calculator	What it does
Sample Size — 1-Sample Mean	Determines the sample size needed to detect a specified difference from a hypothesized mean, given σ, alpha, and target power.
Sample Size — Difference of Two Means	Determines the per-group sample size needed to detect a specified difference between two independent means, given a common σ, alpha, and target power.
Sample Size — 1-Sample Proportion	Determines the sample size needed to detect a difference between a hypothesized proportion and an alternative, given alpha and target power.
Sample Size — Difference of Two Proportions	Determines the per-group sample size needed to detect a difference between two independent proportions, given alpha and target power.
Design Effect & Effective Sample Size	Converts an intraclass correlation (ICC) and average cluster size into a design effect, then discounts a naive total sample size down to its effective (independent-information-equivalent) size — the correction a naive analysis of clustered or nested data misses.
Sample Size — Cluster-Randomized Trial (Two Proportions)	Determines the per-arm sample size and number of clusters needed to detect a difference between two proportions when randomization happens at the cluster level (e.g. clinic, school, or community) rather than the individual level, correcting the standard two-proportion formula with a design effect.
Sample Size — Survival Analysis (Log-Rank / Cox PH)	Determines the number of events — and, given an expected overall event probability, the total sample size — needed for a two-group log-rank test or Cox proportional-hazards model to detect a specified hazard ratio, using Schoenfeld's formula. Unlike the other sample-size calculators here, power depends on the number of events observed, not the number of subjects enrolled.
Sample Size — ANOVA (Cohen's f)	Determines the per-cell sample size needed to detect a specified Cohen's f effect size for a fixed-effects ANOVA main effect or interaction, given alpha and target power — covers both a simple one-way design and a main effect or interaction within a larger factorial design.
Sample Size for a Survey	Determines how many respondents are needed to estimate a population proportion within a target margin of error, with optional finite-population correction and response-rate adjustment.
Power Calculations	Computes statistical power for one- and two-tailed tests from delta, σ, n, and α.
Power with Graph	Visualises the overlap of H_0 and H_a distributions, shading alpha, beta, and power — drag μ_0, μA, σ, n, or α to explore live, with a toggle between one- and two-tailed tests.
Power vs Effect Size & Alpha	Plots power as a function of effect size and significance level for a given sample size.
Post-Hoc Power Calculation	Estimates achieved power for a completed study given its observed effect size and n.
Power, Effect Size, PPV & FPP	Links statistical power to positive predictive value and false positive

Calculator	What it does
	probability.
Power as a Function of δ & α	Shows how power changes across a range of delta and alpha values simultaneously.
Type I & Type II Error Explorer	An interactive decision matrix showing all four possible outcomes of a hypothesis test — two correct decisions and two errors — and how trading off α against power shifts the risk of each.

Diagnostic Testing (6)

Calculator	What it does
Diagnostic Test Accuracy (2×2)	Full diagnostic accuracy calculator: Se, Sp, PPV, NPV, LR+, LR−, and accuracy.
Sensitivity, Specificity & LR	Calculates sensitivity, specificity, and positive/negative likelihood ratios from a 2×2 table.
Post-Test Probability	Updates pre-test probability to post-test probability using likelihood ratios.
ROC Curve & AUC	Plots the receiver operating characteristic curve and computes the area under the curve.
Serial & Parallel Testing	Models the combined performance of tests run in series or parallel.
PPV/NPV vs Prevalence	Shows how positive and negative predictive value change with disease prevalence, for a test with fixed sensitivity and specificity.

ANOVA (12)

Calculator	What it does
1-Way ANOVA	Analysis of Variance (ANOVA) tests for differences in means across three or more independent groups.
Tukey's HSD Test	Post-hoc pairwise comparisons controlling family-wise error rate after ANOVA.
Levene's Test (Brown-Forsythe)	Tests whether two or more groups have equal variances — the precondition check for a standard (pooled-variance) ANOVA. For the two-group case, current guidance is to skip this test and simply use Welch's t-test by default instead.
2-Way ANOVA with Replication	A two-factor Analysis of Variance (ANOVA) that tests main effects and interaction for two factors with multiple observations per cell.
Multi-Factor ANOVA	Extends one-way Analysis of Variance (ANOVA) to designs with more than two independent factors.
MANOVA (2 Outcomes)	Tests whether k independent groups differ on two correlated outcome variables analyzed jointly, using Wilks' Lambda and Pillai's Trace — the multivariate extension of 1-Way ANOVA to two outcomes at once.
Repeated Measures ANOVA	A within-subjects Analysis of Variance (ANOVA) that analyzes data from designs where the same subjects are measured under multiple conditions.

Calculator	What it does
Mauchly's Test of Sphericity	Tests whether the variances of the differences between every pair of repeated-measures conditions are equal — the sphericity assumption a standard Repeated Measures ANOVA depends on.
Holm-Šídák Test	Post-hoc pairwise comparisons following ANOVA, using pooled-variance t-tests with the Holm-Šídák step-down correction for multiple comparisons.
Multiple Comparisons Correction	Applies Bonferroni, Holm, and Benjamini-Hochberg (FDR) corrections to a set of p-values you enter directly — lets you independently check whether a paper's stated multiplicity correction actually holds up.
Shapiro-Wilk Test	Tests whether a sample of continuous data is consistent with a normal distribution — the assumption behind parametric tests like the t-test and ANOVA.
ANCOVA (2-Group, One Covariate)	Analysis of Covariance (ANCOVA) compares two groups' outcome means while statistically adjusting for a continuous baseline covariate — the standard analysis for a two-arm trial with a baseline measurement, and generally more powerful than comparing raw post-scores or change scores alone.

Survival Analysis (3)

Calculator	What it does
Kaplan-Meier Survival Curve	Estimates the probability of surviving past each time point from time-to-event data with censoring, plotting a step curve and reporting median survival time.
Log-Rank Test	Compares survival distributions between two groups using the log-rank test, from time-to-event data with censoring.
Cox Proportional Hazards (Hazard Ratio)	Estimates the hazard ratio between two groups from time-to-event data with censoring, via a univariate Cox regression.

Patient-Reported Outcomes (1)

Calculator	What it does
Minimal Important Difference (MID)	Estimates a minimal important difference (MID) for a patient-reported outcome measure using distribution-based methods (standard error of measurement, half a standard deviation, Cohen's d thresholds), alongside an optional anchor-based estimate for comparison.

Bayesian & Meta-Analysis (9)

Calculator	What it does
Meta-Analysis (Q, τ^2, I^2, PI)	Pools effect sizes, tests heterogeneity, and computes prediction intervals.
Egger's Test	Tests for funnel-plot asymmetry — a regression-based signal of possible small-study effects or publication bias in a meta-analysis.
Bayes' Theorem	Computes posterior probability from prior, likelihood, and marginal

Calculator	What it does
	probability.
Bayesian Credible Intervals	Derives Beta-posterior credible intervals for a proportion given prior and observed data.
Bayes Factor	Quantifies the relative evidence for H_0 vs H_1 on a continuous scale.
Network Meta-Analysis (Indirect & Mixed Comparisons)	Combines direct and indirect evidence across three or more named treatments into one connected network, producing a network diagram, a full league table of pairwise estimates, heterogeneity/inconsistency statistics, and a frequentist treatment ranking (P-scores).
Meta-Analysis for Proportions	Pools proportions (e.g., prevalence or event rates) across studies from raw event counts and sample sizes, using a variance-stabilizing transformation (or a one-stage GLMM), then tests heterogeneity and computes a prediction interval.
Hartung-Knapp-Sidik-Jonkman (HKSJ) Method	Compares the standard (DerSimonian-Laird, normal-based) random-effects confidence interval to the Hartung-Knapp-Sidik-Jonkman adjustment — a refined variance estimate combined with a t-distribution reference — which is typically wider and more robust, especially with few studies or high heterogeneity.
Meta-Analysis for Correlations	Pools correlation coefficients across studies using the Fisher z transformation, tests heterogeneity, and computes a prediction interval for the true correlation.

Genetics & Genomics (2)

Calculator	What it does
Hardy-Weinberg Equilibrium Test	Tests whether observed genotype counts for a biallelic marker match Hardy-Weinberg equilibrium expectations, and reports allele frequencies and the inbreeding coefficient.
Mendelian Randomization (IVW & MR-Egger)	Estimates a modifiable exposure's causal effect on an outcome from genetic instruments (SNPs), using inverse-variance-weighted (IVW) pooling and MR-Egger regression, with instrument-strength (F-statistic) and directional-pleiotropy (Egger intercept) checks.

Probability & Distributions (13)

Calculator	What it does
Binomial Probability Calculator	Computes exact and cumulative binomial probabilities for given n, k, and p.
Critical Value & p-Value (t)	Returns the critical t-value, two-tailed p-value, and implied standard error for a given t statistic and degrees of freedom.
Critical Value & p-Value (z)	Returns the critical z-value, two-tailed p-value, and implied standard error for a given z statistic.
z-Distribution Table	Looks up cumulative probabilities and critical values for the standard normal distribution, with a shaded one- or two-tailed rejection region drawn on the

Calculator	What it does
	curve itself.
t-Distribution Table	Returns critical t-values for one- and two-tailed tests at any df and alpha.
z-Score & Standard Deviation Explorer	An interactive bell curve — drag the raw score, mean, or standard deviation and watch the z-score and percentile recompute live, showing directly how the SAME raw score can be many or few SDs from the mean depending on how spread out the population is. Optionally target an exact z-score directly, or shade a one-/two-tailed rejection region at a chosen α.
t-Distribution Explorer (df & Critical Region)	Draws the actual t-distribution curve at any degrees of freedom, alongside a reference standard normal curve, with the one- or two-tailed critical region shaded — shows directly why a t critical value is more extreme than z at low df, and why the two converge as df grows.
Inverse Probability	Finds the smallest count k such that the binomial cumulative probability P(X ≤ k) meets or exceeds a target probability, given n and p (the BINOM.INV equivalent).
Law of Large Numbers — Coin-Flip Simulator	Simulates repeated coin flips and plots the running proportion of heads against the number of tosses, showing convergence to the true probability as the sample size grows — and how unstable that proportion can be with too few tosses.
Confidence Interval Coverage Simulator	Draws many independent samples and their confidence intervals from a population with a known true mean, showing what fraction of those intervals actually capture it.
Central Limit Theorem Simulator	Draws repeated samples from a chosen non-normal population and plots the growing distribution of their sample means, showing it turn approximately normal as sample size grows — regardless of the population's own shape.
Regression to the Mean Simulator	Simulates two independent measurements of the same trait and shows how a group selected for extreme first-measurement scores drifts back toward the population mean on remeasurement, even though nothing about them actually changed.
Poisson & Negative Binomial	Calculates Poisson and negative binomial probabilities and cumulative distributions.

Non-Parametric Tests (9)

Calculator	What it does
Mann-Whitney U Test	Non-parametric test comparing two independent groups on ordinal or non-normal data.
Wilcoxon Signed-Rank Test	Non-parametric alternative to the paired t-test for comparing two related samples.
Sign Test	Tests the median of paired differences using only the signs of the differences.
Kruskal-Wallis Test	Non-parametric one-way ANOVA for comparing three or more independent groups.
Dunn's Test	Post-hoc pairwise comparisons following a significant Kruskal-Wallis test.

Calculator	What it does
Friedman Test	Non-parametric repeated measures test for comparing three or more related groups.
Aligned Rank Transform (ART) ANOVA	Nonparametric alternative to a 2-way factorial Analysis of Variance (ANOVA), testing both main effects and their interaction when normality or equal-variance assumptions are violated.
Kolmogorov-Smirnov Test (One-Sample)	Tests whether a sample is consistent with a fully specified Normal(μ_0, σ_0) distribution, comparing the sample's empirical CDF against the hypothesized one.
Kolmogorov-Smirnov Test (Two-Sample)	Non-parametric test comparing the full distributions of two independent samples — sensitive to differences in location, spread, or shape, not just central tendency.

Learn Hub: Guide Index

The Learn Hub holds 77 guides across 7 categories — reference material for recognizing data types, reading the charts the calculators produce, and critically appraising the studies you're applying them to. Open the Learn Hub from the "Learn" tab in the hub toggle at the top of any Calculators, Learn, or Designs page, or from the sidebar search.

Data Types (8)

Guide	What it covers
Nominal Data	Categories with no inherent order — the base case for categorical data, and the tests built for it.
Ordinal Data	Categories with a meaningful order, but no fixed distance between them.
Likert Scales: Single Item vs. Summated Scale	A single Likert-type item and a multi-item Likert scale look alike but call for entirely different analyses — one is ordinal, the other is conventionally treated as continuous.
Binary / Dichotomous Data	The two-category special case that shows up constantly in medicine — and unlocks its own set of tools (risk ratios, odds ratios, proportions).
Discrete Count Data	Whole-number counts of events, often over time or exposure — why they need Poisson-style models instead of a plain mean and SD.
Continuous Data (Interval & Ratio)	The most flexible level of measurement, and the one behind t-tests, ANOVA, correlation, and regression.
Paired/Matched vs. Independent Samples	Not a data type — a study-design distinction that determines which half of nearly every test family you need.
Time-to-Event (Survival) Data	Data with a censoring problem: some subjects never experience the event during follow-up, and that has to be handled, not ignored.

Reading and Understanding Graphs (12)

Guide	What it covers
How to Read a Forest Plot	What the squares, lines, and diamonds mean — plus fixed vs. random effects, weights, and prediction intervals.
How to Read a Flow Diagram (Sankey Diagram)	What the bars and ribbons mean, and how one population splits step by step into a 2×2 table.
How to Read a Network Meta-Analysis Diagram & League Table	What the nodes, edges, and league-table cells actually represent, and how direct evidence differs from indirect.
How to Read a Box Plot	What the box, whiskers, and open circles represent, and how the 1.5×IQR outlier rule works.
How to Read a Bland-Altman Plot	Why this plot is about agreement, not correlation, and how to read the bias line and limits of agreement.
How to Read a Baujat Plot	Spotting which studies in a meta-analysis are both discordant and consequential — not just one or the other.
How to Read a Funnel Plot	Why a symmetric inverted funnel is reassuring, and what a gap in one corner might be hiding.
How to Read an Equivalence / Non-Inferiority Zone Plot	The one rule that decides equivalence, and why a non-significant p-value is not the same thing.
How to Read Kaplan-Meier Curves & Cox Proportional Hazards Output	What the step-drops and censoring ticks mean, how the median survival time is read off the curve, and how the Cox hazard ratio quantifies what the Log-Rank Test only tells you yes/no.
How to Read a Fagan Nomogram (and Interpret Likelihood Ratios)	A likelihood ratio only ever multiplies odds, never probability directly — the Fagan nomogram is a straightedge that does that multiplication for you, in one line, without a calculator.
How to Read a Normal Distribution (Bell Curve)	The single most common chart shape in statistics — and the shape quietly underneath nearly every z-score, p-value, and confidence interval calculated on this site.
How to Read a Directed Acyclic Graph (DAG)	A DAG makes your assumptions about what causes what explicit and arguable — and exposes a rule that catches even experienced researchers off guard: adjusting for the wrong variable can create bias out of nothing.

Critical Appraisal of the Literature (19)

Guide	What it covers
How Instrumental Variable Analysis Works (and What Makes a Valid Instrument)	An instrumental variable is a stand-in for randomization — but only if it satisfies three assumptions, two of which can never be directly tested from the data at hand.
Understanding Effect Measures: Relative Risk, Odds Ratio, and Absolute Effects	Relative risk (RR) and odds ratio (OR) answer subtly different questions — and confusing them can meaningfully inflate how big an effect looks.
Mean Difference vs. Standardized Mean Difference: When Units Matter	Two ways to report the same gap between two group means — one keeps the outcome's own units, the other rescales it into SD units so results

Guide	What it covers
and When They Don't	measured on different scales can be compared or pooled.
Appraising Patient-Reported Outcome Measures (PROMs): Validity, Reliability, Responsiveness, and MID	A PROM can be perfectly reliable and still useless for detecting real change — reliability and responsiveness are different properties, and both have to be checked separately.
Appraising Questionnaire and Survey Research: Response Biases, Non-Response, and Choosing the Right Analysis	A well-validated instrument still produces bad data if respondents shade their answers toward what looks good, agree with everything regardless of content, or never receive the questionnaire in the first place.
What a P-Value Actually Means	It is not the probability that the null hypothesis is true, "by chance alone" has a narrower meaning than it sounds like, and 0.05 is a convention, not a law of nature.
One-Tailed vs. Two-Tailed Tests, and When Multiple Comparisons Change the Rules	Two decisions made before a single participant is enrolled — how many directions a test can detect an effect in, and how many hypotheses are being tested at once — quietly change what "significant" means, and both are easy to get right on paper and wrong in spirit.
Confidence Intervals: What "95%" Actually Covers	A 95% confidence interval (CI) is not "a 95% probability the true value lies in this range" — here is the distinction, and why it matters less in practice than you might think.
Confidence Interval, Credible Interval, or Prediction Interval? Three Different Questions	These three intervals look alike on the page — a number, a dash, another number — but they answer genuinely different questions, and mixing them up leads to real misinterpretation.
Frequentist vs. Bayesian Reasoning: Two Ways to Read the Same Evidence	A p-value and a Bayesian posterior can be computed from the exact same data and still answer two genuinely different questions — and clinicians already reason the Bayesian way every time they combine a test result with how likely a diagnosis was beforehand.
Standard Deviation vs. Standard Error: What Error Bars Actually Show	Two quantities that look interchangeable on a figure — a bar, a cap, a number after a "±" — answer genuinely different questions, and mixing them up can make a result look far more precise, or far more separated, than it really is.
Statistical Significance vs. Clinical Significance	"Significant" answers "is this probably not chance?" It does not answer "does this matter to a patient?"
Power, Type II Error, and Why "No Difference Found" Isn't the Same as "No Difference"	An underpowered trial that finds no significant difference has not proven the treatments are equivalent — it may simply have been too small to detect a real effect.
Confounding, Bias, and Why Randomization Matters	An observed association in a cohort or case-control study can come from the exposure being studied — or from something else entirely.
How to Interpret an ANCOVA-Adjusted Result	What "adjusted for baseline" actually buys a trial, and the covariate pitfalls worth checking before trusting the adjusted number.
Homogeneity of Variance and Sphericity: The Constant-Variance Assumptions Behind ANOVA-Family Tests	Two precondition checks with unrelated-sounding names that are really the same question — does spread stay constant? — asked of two different data shapes: independent groups, and repeated measurements on the same subjects.
Study Design Hierarchy and Its	Randomized controlled trial (RCT) > cohort study > case-control study > case

Guide	What it covers
Exceptions	series is a reasonable default — but a poorly run RCT can be less trustworthy than a well-run cohort study.
Diagnostic Tests Depend on Prevalence	Sensitivity and specificity are properties of the test itself. Positive predictive value (PPV) and negative predictive value (NPV) are properties of both the test and the population it is used in.
Survival Analysis Basics: Censoring, Hazard Ratios, and Proportional Hazards	A hazard ratio (HR) describes relative risk over time — not the absolute difference in how long people lived.

Evidence Synthesis & Certainty (4)

Guide	What it covers
Reading a Meta-Analysis: Beyond the Pooled Estimate	The pooled diamond on a forest plot is only as trustworthy as the heterogeneity behind it — and pooling studies cannot fix flawed primary research.
Reporting Guideline, Risk-of-Bias Tool, or Certainty Framework? Four Different Jobs	CONSORT, RoB 2, AMSTAR-2, and GRADE all get invoked in the same breath, but they check four completely different things — and a study can pass one perfectly while failing another.
Understanding GRADE: Certainty of Evidence vs. Strength of Recommendation	GRADE rates two different things that are easy to collapse into one — how much to trust the evidence, and how strongly to act on it — and conflating them is one of the most common misreadings of a systematic review or guideline.
When a Paper's Conclusions Outrun Its Own GRADE Rating	A systematic review can rate an outcome "very low certainty" in its own GRADE table and still describe that same outcome in flatly confident language a few pages later in the Conclusions — a mismatch worth checking for by name, not assumed away.

Common Statistical Pitfalls (8)

Guide	What it covers
Regression to the Mean	Why a before/after study with no control group will almost always show "improvement" — treatment or not.
Reading a Monte Carlo (Simulated) p-value	A permutation-based p-value has a precision floor set by how many simulations were run — "p < 0.001" from 1,000 permutations isn't what it looks like.
The Ecological Fallacy	A country, school, or hospital can show one pattern in its aggregate numbers while every individual inside it shows the opposite — and confusing the two is one of the most common, and most avoidable, misreadings in the literature.
Subgroup Analyses and Interaction Tests	“The treatment worked in women but not men” is one of the most common misreadings in the clinical literature — and usually isn’t what the data actually show.
Sensitivity Analysis vs. Subgroup	"We checked robustness" can describe two very different exercises — and

Guide	What it covers
Analysis	mixing them up hides two very different risks.
Too Good to Be True: Recognizing Overinterpreted Study Results	"Significant" and "groundbreaking" are two different claims — and a paper's discussion section is where they're most often blurred together.
"Post Hoc" Test, Power, or Analysis? Three Different Things Sharing One Name	The same two Latin words describe a rigorous, built-in correction procedure, a circular power calculation, and the single biggest source of manufactured "significant" findings in the literature — a paper rarely says which one it means.
The Table 2 Fallacy	A multivariable model is built to give one unconfounded estimate — the exposure of interest. The other coefficients sitting right next to it in the same table are not free bonus results.

Appraising Studies by Design (18)

Guide	What it covers
Appraising Randomized Controlled Trials	Randomization solves confounding in principle — whether a specific trial delivers on that promise depends on a handful of design details worth checking every time.
Appraising Cohort Studies	A cohort study follows exposed and unexposed groups forward through time — appraisal here means checking whether those two groups stayed comparable the whole way, not just at the start.
Appraising Case-Control Studies	A case-control study works backward from outcome to exposure — which means its appraisal concerns center on how cases and controls were chosen and how honestly the past can actually be reconstructed.
Appraising Cross-Sectional Studies	A cross-sectional study captures exposure and outcome at the same instant, in the same subjects — which means its biggest appraisal question is one no other design shares: which came first?
Appraising Systematic Reviews and Meta-Analyses	A systematic review is only as trustworthy as the individual studies it pools — appraisal here means checking both the review's own conduct and its handling of the evidence underneath it.
Appraising Scoping Reviews	A scoping review answers "what has been studied, and how" — not "what works" — and appraising one means checking whether it stayed true to that different purpose.
Appraising Realist Reviews	A realist review asks "what works, for whom, under what circumstances, and why" — not "does it work" — and appraising one means checking whether it explains mechanisms rather than just pooling outcomes.
Appraising Diagnostic Accuracy Studies	Sensitivity and specificity look like fixed properties of a test — but how a study selected its patients and defined its reference standard can shift both numbers substantially.
Appraising AI/ML Diagnostic and Prediction Studies	An AI model's reported accuracy is only as trustworthy as how it was trained, validated, and tested — and that process introduces failure modes a conventional diagnostic accuracy study doesn't have to worry about.
Appraising Pilot and Feasibility Studies	Pilot studies are almost never adequately powered to detect the effect they may report — reading one as if it were is one of the most common

Guide	What it covers
	misreadings in the literature.
Appraising Prognostic Studies	A prognostic study asks what happens next, not whether a treatment works — which shifts the appraisal questions toward how the cohort was assembled and how honestly its follow-up data were handled.
Appraising Clinical Practice Guidelines	A guideline is a step removed from any single study — appraising one means checking not the evidence itself, but whether the group that weighed it did so completely and transparently.
Appraising Cluster-Randomized Trials	Randomizing clinics, schools, or villages instead of patients solves a practical problem — but it creates a statistical one an ordinary trial never has to deal with.
Appraising Crossover Trials	Each patient serves as their own control, which is powerful — but only if enough of the first treatment washes out before the second one starts.
Appraising Non-Inferiority Trials	A poorly conducted non-inferiority trial is biased toward the very conclusion it's trying to prove — the opposite problem a sloppy superiority trial has.
Appraising Ecological Studies	Ecological studies compare groups, not individuals — and that shift in the unit of analysis is exactly where the ecological fallacy comes from.
Appraising Case Series and Case Reports	The simplest design in the hierarchy — no comparison group at all — which is exactly why a case series can describe but never truly test anything.
Appraising Genetic Association Studies: Candidate Gene, GWAS, and Mendelian Randomization	Candidate gene studies built a decades-long replication crisis on underpowered, uncorrected testing. The Genome-Wide Association Study (GWAS) fixed that with genome-wide significance and mandatory replication — and Mendelian Randomization goes further, using genotype itself as nature's own randomized experiment to make causal claims from purely observational data.

Quick Reference (8)

Guide	What it covers
The Biostat Toolkit — User Guide (Downloadable PDF)	A downloadable PDF walkthrough of the site as a whole — what it is, how the calculators, Learn Hub, and decision wizards fit together, and how to get the most out of it offline, in print, or as a citable handout.
Glossary of Statistical Abbreviations and Symbols	A quick lookup for the abbreviations and symbols used throughout this site — grouped by topic, with links to the fuller guide or calculator where one exists.
Glossary of Statistical Concepts and Roles	A quick lookup for the ordinary-sounding words statistics gives a specific technical meaning to — variable, covariate, concordant/discordant, and more — grouped by topic, with links to the fuller guide or calculator where one exists.
Hypothesis Testing: A Quick Reference	The logic every hypothesis test follows, one-tailed vs. two-tailed, the decision rule, and a lookup table for which test statistic applies to which situation.
Minimal Important Differences for Common PRO Instruments	Published MID/MCID reference values for five widely used patient-reported outcome instruments, each with its own citation and caveat — a starting

Guide	What it covers
	lookup, not a substitute for checking the primary source.
Critical Appraisal Worksheets (Downloadable)	Six fill-in-the-blank worksheets — one per clinical question type — for critiquing a specific article. Download the one matching the article you're reading and work through it question by question.
Visual Trade-off Tools on This Site	An index of every interactive chart on this site that shows a quantity trading off against another as you move a parameter — sensitivity vs. specificity, Type I vs. Type II error, and more.
Live Simulators on This Site	An index of every calculator on this site that runs a live random simulation in your browser — repeated coin flips, repeated samples, repeated confidence intervals — rather than computing a single result from numbers you enter.

Study Designs Hub

For planning a study rather than analyzing one already in hand, the Designs hub is a quick-reference map from a research question to the design built for it. Seventeen designs are organized into five groups; each design card links straight to its own in-depth critical-appraisal guide in the Learn Hub.

Experimental Trials (4)

- Randomized Controlled Trials
- Cluster-Randomized Trials
- Crossover Trials
- Non-Inferiority Trials

Observational — Comparative (2)

- Cohort Studies
- Case-Control Studies

Observational — Descriptive (3)

- Case Series and Case Reports
- Cross-Sectional Studies
- Ecological Studies

Diagnostic & Prognostic (3)

- Diagnostic Accuracy Studies
- AI/ML Diagnostic and Prediction Studies
- Prognostic Studies

Evidence Synthesis & Planning (5)

- Systematic Reviews and Meta-Analyses
- Scoping Reviews

- Realist Reviews
- Clinical Practice Guidelines
- Pilot and Feasibility Studies

Cross-Sectional Studies and Ecological Studies are grouped here under "Observational — Descriptive" alongside Case Series and Case Reports, reflecting how the Study Design Finder wizard itself branches (all three answer "how common is this, or what happened in this group" rather than a forward/backward exposure-to-outcome comparison).

Genetic Association Studies (candidate gene, GWAS, and Mendelian Randomization) has its own in-depth guide in the Learn Hub's "Appraising Studies by Design" category, but does not currently have a matching card or wizard branch in the Designs hub itself — reach it directly from the Learn Hub rather than the Study Design Finder.

Every other design listed here corresponds one-to-one with an "Appraising ..." guide in the Learn Hub's "Appraising Studies by Design" category — the two are not written separately, so they don't drift apart.

8. Practical Tips

- Nothing you type is saved between visits or sent anywhere — every calculation runs locally in your browser. Copy or export results before closing the tab.
- If you're not sure whether a guide or a calculator is the right starting point, the Designs hub is usually the fastest route when you're still planning a study, and the Calculator Finder is fastest once you have data in hand.
- The Learn Hub's "Quick Reference" category holds the glossary, downloadable critical-appraisal worksheets, a downloadable PDF copy of this User Guide, and an index of every visual trade-off tool and live simulator on the site — worth bookmarking.
- On phones and narrow windows, the sidebar collapses behind a menu button (☰) in the top corner; tap it to reveal search and navigation.

9. About This Toolkit

The Biostat Toolkit is developed and maintained as an educational resource. For a related set of short illustrated vignettes on statistical concepts, see "Stats with Crayons" from Penn Dental Medicine's Center for Integrative Global Oral Health, linked from the Learn Hub's Quick Reference section.

This guide reflects the site's contents as of the date on the cover. As calculators and guides are added, the counts and category lists above may fall slightly out of date — the live counts shown on the site itself (top of each hub page) are always authoritative.